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Superconductivity in transparent amorphous indium tin oxide films deposited by RF magnetron sputtering

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Integrating superconductivity with high optical transparency is critical for advancing quantum technologies, yet remains fundamentally challenging due to photon damping at conventional superconductor interfaces. Here, we report the integrated circuit process compatible and scalable fabrication of transparent superconducting tin-doped indium oxide (ITO) thin films by RF magnetron sputtering without post-treatment. Combining Aslamazov-Larkin fluctuation theory and Ginzburg-Landau analysis, a superconducting transition temperature T_c of 1.43 K and the zero-temperature coherence length of 18.14 nm are determined. The two-dimensional nature of superconductivity is corroborated by a Berezinskii-Kosterlitz-Thouless transition near 0.8 K. These ITO films exhibit high transmittance across visible and near-infrared wavelengths, meeting the demands of quantum photonic applications. Comparative magneto-transport studies reveal that disorder suppresses electron-phonon coupling, thereby reducing T_c or even quenching superconductivity, while superconductivity itself suppresses weak localization (WL) signatures. Based on these observations, we propose a four-stage model describing the evolution from a normal metallic state to a phase-coherent superconducting phase through an interactive regime and a pre-pairing fluctuation regime. This work provides a simple and easy growth method for transparent ITO superconducting films and paves the way of exploring transparent superconductor for promising quantum material platforms.

Transparent superconductors—materials that exhibit both zero electrical resistance and high optical transmittance—offer a compelling platform for integrating quantum computation and photonics[1,2]. In particular, they are considered ideal candidates for enabling coherent microwave–optical signal conversion, a critical component for scalable quantum networks[1,3].

However, achieving transparent superconductivity remains challenging, as the requirements for high electrical conductivity and optical transparency are inherently at odds. Among the limited systems explored, indium oxide-based materials such as amorphous InO_x and tin-doped indium oxide (ITO) have demonstrated potential. Early studies primarily investigated disorder-driven superconductor–insulator transitions (SIT) in InO_x films prepared by reactive ion-beam sputtering[4,5]. Around the same time, Mori et al.. reported superconductivity in ITO films via post-annealing after electron-beam deposition[6,7]. Yet despite these early efforts, reproducible and controllable methods to achieve superconducting InO_x or ITO films remain elusive[8,9]. Film properties are highly sensitive to deposition parameters such as pressure, substrate temperature, and ambient atmosphere, and no clear predictive criteria for superconductivity in transparent oxides have been established. More recently, electrochemical doping has been used to induce superconductivity in ITO films[10,11]. While effective in triggering superconducting transitions, this method introduces surface defects and severely compromises optical transparency in the visible–infrared range. Moreover, such chemical treatments are incompatible with standard integrated circuit (IC) fabrication workflows used in superconducting quantum devices. These limitations underscore the urgent need for a simple, IC-compatible, and reliable approach to grow transparent superconducting films.

Here, we report a scalable and IC-compatible sputtering approach to fabricate centimeter-scale amorphous ITO thin films with tunable disorder and controllable superconductivity, without requiring post-processing.

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We demonstrate that reducing disorder—quantified by the ratio of the mean free path to Fermi wavelength—is key to enabling superconductivity. Furthermore, magneto-transport analysis reveals an unusual deviation from the expected phase coherence behavior far above the superconducting transition temperature, suggesting the presence of an intermediate interaction-dominated regime. These findings not only demonstrate an effective route to induce superconductivity in transparent ITO films and establish a platform for exploring superconductor–photon interactions, but also hold promise for advancing hardware development in quantum computing networks.

Results

Manipulation of electrical and optical properties

The electrical and optical properties of ITO thin films strongly depend on the deposition method and conditions, including techniques such as PLD, ALD, spin-coating, and sputtering[12]. In this work, we fabricated transparent ITO films via reactive RF magnetron sputtering. A systematic set of growth parameters was explored, and the detailed fabrication conditions together with sample statistics are provided in the Supplementary Information. In total, more than twenty films (thicknesses < 50 nm) were examined, from which clear trends emerge regarding the interplay between deposition conditions, structural characteristics, and superconductivity. The key variables are oxygen partial pressure and substrate temperature: superconductivity is only observed in films grown under relatively low oxygen partial pressure (< 2%) and low substrate temperature (< 160 °C), where amorphous structures are favored. In contrast, higher oxygen partial pressures (> 2%) and elevated substrate temperatures (> 160 °C) promote crystallization and suppress superconductivity.

For clarity in the main text, two representative samples are highlighted. Sample 1 (S1), grown at 1.98% oxygen partial pressure and 140 °C, exhibits an amorphous structure and shows a superconducting transition. By contrast, Sample 2 (S2), deposited at 20% oxygen partial pressure and 240 °C, is crystalline and does not exhibit superconductivity. These two samples illustrate the broader trends confirmed by the full set of sample statistics presented in the Supplementary Information (Figure S1 and Table 1).

In two-dimensional (2D) systems, the degree of disorder is commonly evaluated based on resistivity, quantum interference corrections to conductivity and the characteristic temperature at which these corrections become appreciable[13]. Typically, as shown in Fig. 1a, films with higher sheet resistance $R_{\square}$ exhibit stronger disorder[13,14]. which enhances electron scattering and the quantum interference between scattering paths and leads to the additional sheet resistance increase upon further cooling. In weakly disordered systems (S1), lower temperatures are generally required to observe weak localization.

Superconductivity was induced by optimizing deposition parameters to suppress disorder, as demonstrated by the transition in transport properties from S2 to S1. As shown in Fig. 1b, films with higher crystallinity tend to exhibit a higher degree of disorder, whereas amorphous films typically possess relatively lower disorder. S1 was amorphous, indicated by the absence of diffraction peaks, while S2 exhibited a preferred ⟨111⟩ growth orientation with a (222) diffraction peak appearing at $2\theta = 30.17°$[7,15]. Both structure and transport properties can serve as convenient criteria for tunning deposition parameters to achieve superconducting films.

Potential applications in quantum technologies require ITO superconducting films to maintain high transparency over a broad spectral range, particularly within the telecommunications window. Figure 1c shows the transmittance spectra of S1' and S2', which were deposited simultaneously with S1 and S2 onto clean BaF_2 crystal substrates. Both films exhibit high transmittance (> 70%) in the visible (380–780 nm) and near-infrared (780–2500 nm) regions. The absorption edges in the ultraviolet region (200–380 nm) correspond to the optical

Sample No.	Temperature ℃	O_2 partial pressure %	Chamber pressure Pa	RF Power W	Signature of superconductivity?
1	200	0	0.31 ± 0.01	350	Yes
2	140	0	0.35 ± 0.01	380	Yes
3	140	1.98		380	Yes
4	140	4.35		380	No
5	150	0		380	Yes
6	150	1.98		380	Yes
7	150	4.35		380	No
8	160	0		380	Yes
9	160	1.98		380	Yes
10	160	4.35		380	No
11	180	0		380	
12	180	1.98		380	
13	180	4.35		380	
14	200	0		380	Yes
15	200	1.98		380	No
16	200	4.35		380	No
17	240	20	0.41 ± 0.01	380	No

Table 1. ITO films grown under different deposition parameters.

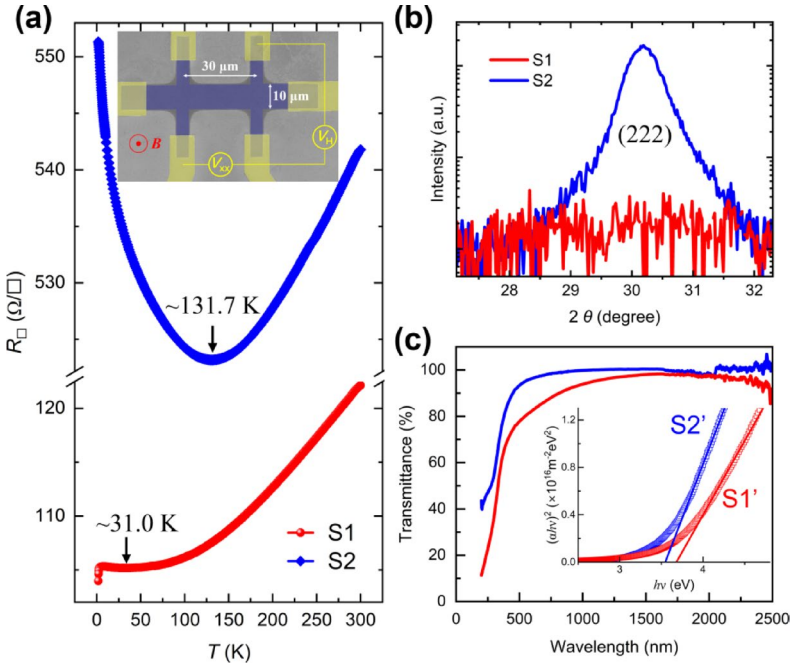


Fig. 1. Electrical, structural and optical properties of superconducting and normal metallic ITO thin films. **(a)** Sheet resistance $R_\square(T)$ as a function of temperature. Inset: False-colored SEM image of a representative Hall device. Blue part is the ITO channel and yellow parts are Ni/Au electrodes. **(b)** X-ray diffraction patterns showing a preferred (222) orientation. **(c)** Optical transmittance spectra. Inset: Bandgap estimation using Tauc plot extrapolation.

bandgap energy (E_g), which was evaluated by extrapolating the linear portion of the $(\alpha h\nu)^2$ versus $h\nu$ plot[16,17]. The extracted optical band gaps are 3.68 eV for S1' and 3.55 eV for S2'. The lower transmittance of superconducting film at wavelengths below ~ 1000 nm is likely due to enhanced free carrier absorption resulting from increased carrier concentration[18], a subject currently under further investigation. For potential wave guide integration in quantum chips, the refractive index (n) and extinction coefficient (k) extracted via spectroscopic ellipsometry are provided in the supplementary material (Fig. S2).

Superconductivity and upper critical fields

Next, basic superconducting properties of transparent ITO thin films are discussed. The temperature-dependent $R_\square$ of S1 under various out-of-plane magnetic fields is shown in Fig. 2a. A zero-resistance superconducting state persists under magnetic fields up to approximately 15 mT, with superconductivity fully suppressed near 1 T. The field-dependent $R_\square$ at temperatures 0.2 to 3 K is presented in Fig. 2b. To accurately construct B–T phase diagram, precise determination of the upper critical field $\mu_0 H_{c2}(T)$ is required. Conventional criteria, such as the midpoint of the resistive transition or linear extrapolation, are unsuitable for disordered 2D systems due to significant broadening induced by superconducting fluctuations[19]. Instead, we employed a self-consistent method: the zero-field $R(T)$ curve near the transition was fitted using the 2D Aslamazov-Larkin (AL) theory to extract the mean-field transition temperature T_c (Fig. S3)[19]. The resistance at $T = T_c$ under zero field, $R(H = 0, T_c)$, was then used as the criterion to identify $\mu_0 H_{c2}(T)$ at lower temperatures (see supplementary material Note II for details). The resulting T_c–$\mu_0 H_{c2}$ data are shown in Fig. 2c. Between 1.2 and 0.5 K, $\mu_0 H_{c2}$ increases linearly with decreasing temperature, consistent with dirty type-II superconductivity. Below 0.5 K, a more rapid, nonlinear increase is observed, which may indicate the onset of quantum Griffiths phase[20], warranting further investigation. To verify $\mu_0 H_{c2}(0)$, the linear portion of the data was fitted using the phenomenological two-dimensional Ginzburg-Landau (GL) theory:

$$H_{c2,\perp}(T) = \frac{\Phi_0}{2\pi \xi_{GL}^2(0)}\left(1 - \frac{T}{T_c}\right), \quad (1)$$

where ξ_{GL} is the in-plane GL coherence length at $T = 0$ K and $\Phi_0 = h/2e$ is the magnetic flux quantum. The fit yields $\mu_0 H_{c2}(0) = 0.33$ T and $\xi_{GL}(0) = 18.14$ nm, with $T_c = 1.43$ K. The close match between the T_c determined via 2D A-L fitting ($T_c = 1.48$ K, see supplementary material) and that extracted from 2D G-L analysis confirms the self-consistent and physically meaningful nature of the procedure, where the AL-based reference resistance serves as a robust criterion for the superconducting transition under magnetic fields.

The two-dimensional nature of superconductivity was further confirmed via Berezinskii-Kosterlitz-Thouless (BKT) transition analysis[21,22]. In the BKT framework, the voltage–current (V–I) characteristics follows a power law $V \sim I^{\alpha(T)}$, where the exponent $\alpha(T)$ is related to superfluid stiffness. The transition occurs at the temperature T_{BKT} where $\alpha(T_{BKT}) = 3$, marking the unbinding of vortex–antivortex pairs. Below T_{BKT} ($\alpha > 3$),

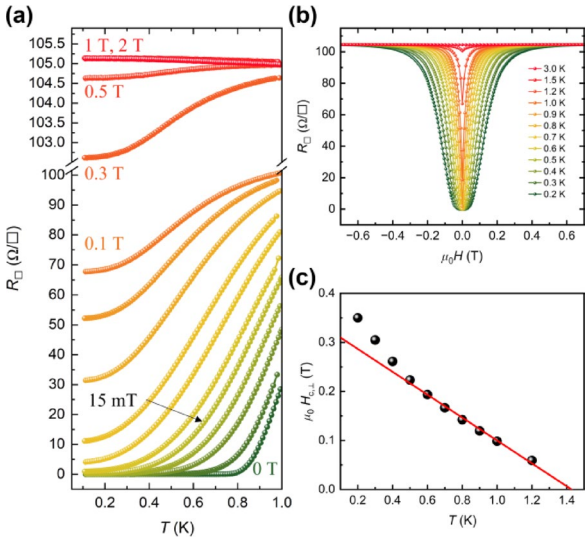


Fig. 2. Superconductivity and upper critical field of transparent ITO thin film. **(a)** Temperature-dependent $R_{\square}$ under perpendicular magnetic fields. **(b)** Field-dependent $R_{\square}$ from 0.2 to 3 K. **(c)** The upper critical field $\mu_0 H_{c,\perp}(T)$, extracted from **(b)**.

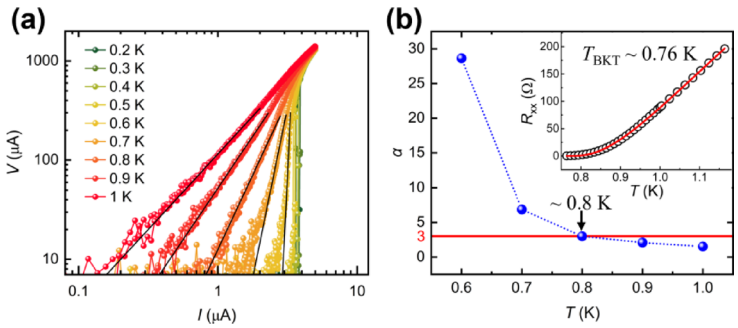


Fig. 3. The BKT transition in S1. **(a)** $V(I)$ characteristics from 0.2 to 1 K on a double-logarithmic scale. fitted by $V \propto I^{\alpha}$. **(b)** Temperature dependence of exponent α. Inset: Resistance vs. temperature fitted by BKT model.

free vortices are frozen into bound pairs, preserving phase coherence and establishing a zero-resistance state. Above T_{BKT} ($\alpha < 3$), vortex pairs dissociate, leading to a finite resistance due to mobile free vortices. The double-logarithmic plots of I–V characteristics at various temperatures are shown in Fig. 3a, from which $\alpha(T)$ was extracted via linear fits. The temperature dependence of $\alpha(T)$ (Fig. 3b) indicates a BKT transition around 0.8 K. Additionally, the temperature dependence of $R(T)$ near T_{BKT} can be described by the Halperin-Nelson formula, $R = R_0 exp\left\{-b\left(\frac{T}{T_{BKT}} - 1\right)^{-1/2}\right\}$, where R_0 and b are material-dependent parameters[23]. The fitting (inset of Fig. 3b) yields $T_{BKT} \approx 0.76$ K, consistent with the value obtained from the I–V analysis.

To further validate superconductivity, we performed magnetization measurements on a thicker (~100 nm) ITO film. As shown in Fig. S4, the onset of a diamagnetic signal is observed around 0.9 K, close to the zero-field superconducting transition temperature defined by the onset of zero resistance (Fig. 2a), featuring a sharp superconducting transition width ($\Delta T < 0.1$ K at 50% diamagnetic moment). Due to the limited thickness, the overall magnetic signal is weak, leading to only minor differences between the zero-field-cooled (ZFC) and field-cooled (FC) magnetization curves.

Correlation between weak localization and superconductivity

The superconductivity was only induced in moderately/weaky disordered amorphous ITO films rather than strongly disordered polycrystalline ITO films. To elucidate the underlying correlation, we examine the characteristic length scales that reflect the nature of charge transport and can be extracted from magnetoconductance analysis[24,25]. The magnetoconductance ΔG, $\Delta G = G(B) - G(0)$, is quantified by using a unit of $e^2/\pi h$, where h is Plank's constant and e is the unit charge. A standard Hikami-Larkin-Nagaoka (HLN) formula, interpreting a 2D diffusive system with low carrier mobility, is expressed as

$$\Delta G(B) = \frac{\beta e^2}{2\pi^2 \hbar}\left[\psi\left(\frac{1}{2}+\frac{B_\varphi}{B}\right) - \ln\left(\frac{B_\varphi}{B}\right)\right], \quad (2)$$

where $B_\varphi = \hbar/4el_\varphi^2$, $\psi(x)$ is the digamma function, B_φ is the dephasing magnetic field determined by the phase coherence length l_φ[24]. The coefficient β represents the total number of conduction channels in the thin film. The fits of both S1 and S2 were perfect and plotted as solid lines in Fig. 4a and b, respectively. For visual clarity, only the data at selected temperatures is shown and the details of the fitting are discussed in supplementary material. The extracted temperature-dependent l_φ and β of S1 and S2 were plotted in Fig. 4c and d, respectively.

The ΔG of S1 and S2 were analyzed over the temperature range of 1.8 to 80 K, where the weak localization effects dominated. Firstly, the extracted coefficients β of both samples are distributed between 0.34 and 0.40. According to the HLN theory, β should approach 1 in the absence of spin-orbit interaction and magnetic scattering[24,26]. The observed suppression of β thus indicates a partial reduction of quantum interference corrections. To explore the origin of this deviation, we employed a modified HLN model to fit the ΔG data and isolate the influence of spin-orbit coupling on quantum interference (see supplementary material Note III for details). The fitting results reveal that the suppressed β is not due to strong spin-orbit interaction; rather, it arises from disorder-induced partial localization, where only extended electronic states contribute effectively to phase-coherent transport. Within this framework, β serves as an effective measure of the fraction of electrons participating in coherent interference. The spin–orbit scattering length in both samples greatly exceeds the phase coherence length (Fig. S5 in supplementary material), confirming the negligible role of spin–orbit coupling—consistent with earlier reports on disordered ITO systems[8,9]. Secondly, l_φ exhibits a power-law dependence on temperature, $l_\varphi \propto T^{-p}$. Power-law fitting yields exponents of 0.58 ± 0.02 for S1 (20–80 K) and 0.54 ± 0.03 for S2 (1.8–80 K). These values are in good agreement with the Nyquist dephasing mechanism[13,27], which predicts $l_\varphi \propto T^{-1/2}$ in 2D system due to inelastic electron–electron (e–e) scattering mediated by low-energy electromagnetic fluctuations. This indicates that e–e interactions dominated dephasing in both films. Given the high electron concentration (Fig. S7) and the strong orbital overlap of *ns* orbitals of the metallic elements in these degenerated ITO films, phonons contributions to decoherence and charge transport are expected to be minimal[28].

Despite the similar value of β and l_φ for samples S1 and S2 in the 20–80 K range—suggesting comparable dephasing mechanisms and conduction channels—these parameters alone cannot account for two notable observations. First, S1 and S2 display markedly different onset temperatures and magnitudes of quantum interference corrections. Second, their low-temperature $l_\varphi(T)$ behaviors diverge: in non-superconducting S2, $l_\varphi(T)$ follows a power-law dependence down to 1.8 K, while in superconducting S1, deviations from this trend

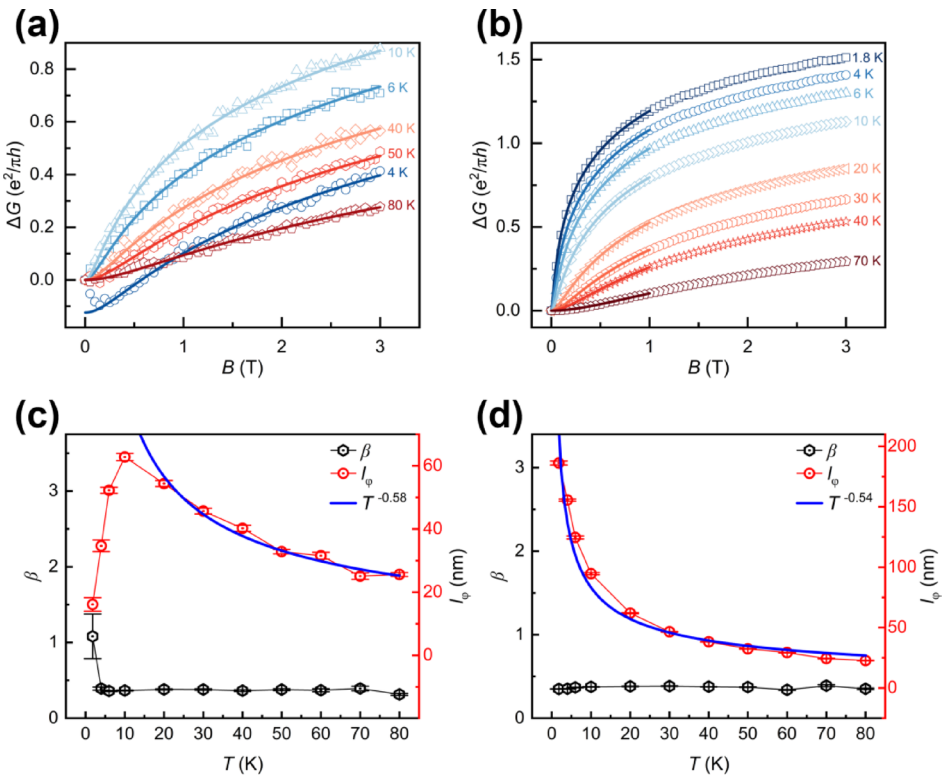


Fig. 4. Quantum interference and dephasing in ITO thin films. **(a, b)** Magnetoconductance ΔG vs. perpendicular field for S1 and S2, with HLN fits (solid lines). **(c, d)** Temperature dependence of phase coherence length l_φ and channel factor β; blue solid lines show $l_\varphi \propto T^{-p}$, with $p = -0.58 \pm 0.02$ (S1) and -0.54 ± 0.03 (S2).

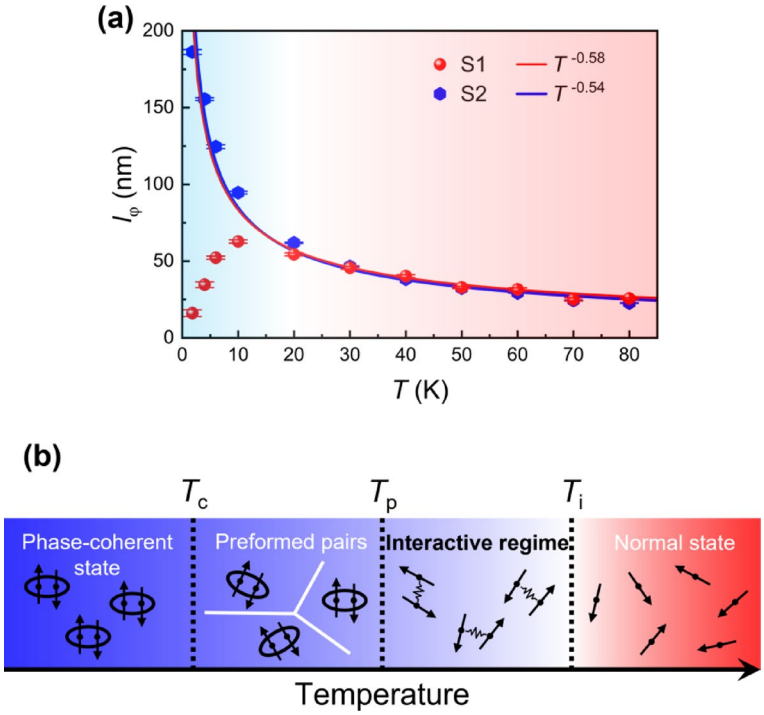


Fig. 5. Competition between superconducting fluctuations and localization. **(a)** Temperature dependence of phase coherence length l_φ, with fits $l_\varphi \propto T^{-p}$, where $p = -0.58 \pm 0.02$ (S1) and -0.54 ± 0.03 (S2). **(b)** Schematic of electronic states: from normal metal to phase-coherent superconductivity via an interactive regime and a pre-pairing regime with incoherent Cooper pairs.

emerge below 20 K. Given that the strength of weak localization is governed not only by l_φ, but also by the degree of disorder and the diffusive transport, the key difference may lie in the mean free paths l of each sample.

The disorder-induced localization strength is quantified by the ratio l/λ_F where λ_F is the Fermi wavelength (Fig. S8)[29]. Both samples show a weak power-law temperature dependence of l, with fitted exponents of − 0.19 (S1) and − 0.09 (S2), indicating that electron transport is primarily limited by temperature-independent elastic scattering from defects, grain boundaries, dislocations, and nonmagnetic impurities[30]. Below approximately 70 K, the mean free path saturates, with S1 maintaining roughly twice the mean free path of S2, consistent with weaker static disorder.

For amorphous S1, $l > \lambda_F$ places it in a weak-scattering regime supportive of metallic diffusion. In contrast, polycrystalline S2, with $l < \lambda_F$, approaches the Ioffe–Regel limit[29], where elastic scattering dominates and wave-like transport collapses. This is reflected in their WL behaviors: S2 exhibits quantum interference corrections up to 130 K, while S1 shows such corrections only below 40 K. Moreover, WL is stronger in S2, consistent with the relation $\Delta G_{2D} \propto -ln\,(l_\phi/l)$, where a smaller l enhances localization effects[31].

The deviation of the $l_\varphi(T)$ from power-law behavior in S1 below 20 K suggests a superconductivity-related effect, as shown in Fig. 5a. According to Anderson's theorem, non-magnetic disorder does not suppress *s*-wave superconductivity provided time-reversal symmetry is preserved[32], but this applies only under weak disorder. In stronger disorder regimes, localization can weaken electron-phonon coupling, suppress pairing, and lower or eliminate the superconducting transition temperature T_c[14,33,34]. Conversely, superconducting correlations can also impact WL: electron pairing disrupts single-electron phase coherence, leading to enhanced dephasing and suppressed l_φ. Thus, the observed suppression of l_φ below 20 K in S1 likely signals the emergence of pairing fluctuations. In S2, where $l < \lambda_F$, strong localization precludes the formation of extended states needed for Cooper pairing, thereby suppressing superconductivity. In contrast, S1 satisfies $l > \lambda_F$, allowing pairing to emerge. While global superconductivity is absent above T_c, the deviation of l_φ from power-law behavior reflects competition between disorder and pairing interactions. Similar precursor superconducting states, featuring incoherent Cooper pairs above T_c, have been observed in InO_x and TiN systems through tunneling and noise spectroscopy[35–37]. Based on spectroscopic results, a three-stage model has been proposed for disordered superconductors: metallic, pre-paired (incoherent Cooper pairs), and fully superconducting states, with a characteristic pre-pairing temperature $T_p \sim 2$–$4\ T_c$[35–37].

In our ITO films, the phase coherence length l_φ exhibits a notable deviation from the conventional Nyquist e–e dephasing behavior at temperatures well above T_c. This deviation emerges around 20 K, where no thermodynamic signatures of a phase transition are observed, suggesting that it originates from an additional dephasing mechanism rather than from pre-formed Cooper pairing. As the temperature decreases below ~ 10 K, l_φ sharply drops, consistent with the onset of superconducting fluctuations. However, in the intermediate temperature window (~ 10–20 K), the magnitude and temperature dependence of the deviation cannot be

easily explained by short-lived Cooper pair fluctuations, which are expected to be negligible far above T_c[38]. We therefore refer to this regime as an "interactive regime", where electron interactions—particularly electron–phonon scattering—significantly enhance decoherence beyond the conventional Nyquist mechanism, bridging the high-temperature normal-state dephasing and low-temperature superconducting-fluctuation-dominated behavior. The onset temperature of "interactive regime" is defined as T_i, schematically illustrated in Fig. 5b.

To further substantiate this interpretation, complementary experimental probes can be employed. Nernst effect measurements can reveal vortex-like excitations and short-lived Cooper pairing above T_c[39]; microwave and optical spectroscopy can identify electron–phonon scattering features in the complex conductivity without the low-frequency reactive response characteristic of superconducting fluctuations[40]; and noise spectroscopy may detect correlated transport processes indicative of inelastic interactions rather than pairing[37]. The combination of these techniques could distinguish an interaction-dominated decoherence regime from a genuine pre-pairing or fluctuation regime, providing indirect but convincing evidence for the existence of the intermediate "interactive regime".

Conclusions

To summarize, we have developed a scalable, integrated-circuit-compatible sputtering method for fabricating centimeter-scale transparent superconducting ITO thin films, without the need for post-deposition treatments. These films exhibit high optical transmittance across the visible to near-infrared spectrum and reproducible superconducting behavior, offering a practical platform for integrating superconducting quantum devices with photonic systems. Systematic analysis reveals that the emergence of superconductivity in quasi-2D ITO films is closely tied to the level of disorder, which can be characterized by the ratio of the mean free path to the Fermi wavelength l/λ_F. When disorder is weak ($l > \lambda_F$), electron pairing can overcome localization, allowing superconductivity to develop. Conversely, strong disorder suppresses T_c and enhances weak localization. In superconducting samples, a notable deviation of the phase coherence length from its expected power-law dependence at low temperatures indicates the onset of preformed Cooper pairs above T_c. Based on these findings, we propose a four-stage evolution from a metallic state to phase-coherent superconductivity, progressing through an intermediate interactive regime and a pre-pairing fluctuation regime. This work not only elucidates the interplay between disorder, coherence, and superconductivity in ITO films, but also repositions ITO as a promising material for transparent superconducting devices in future quantum information networks.

Methods

Transparent ITO thin films were deposited using a commercial RF magnetron sputtering system (ULVAC, Japan). Key deposition parameters, including RF power (380–420 W), oxygen partial pressure, and substrate temperature (120–240 °C), were systematically optimized. Prior to deposition, the chamber was evacuated to 5×10^{-4} Pa and substrates were heated to the target temperature. Argon (sputtering gas) and oxygen (reactive gas) were introduced at a total flow rate of 80 sccm to stabilize the chamber pressure at 0.34 ± 0.01 Pa. Oxygen partial pressure was varied to modulate carrier concentrations. The ITO target (In: Sn = 90:10 by weight, purity > 99.99%, diameter ~ 96 mm) was powered by RF excitation. In our systematic studies, films deposited at substrate temperatures below 180 °C exhibited amorphous characteristics or extremely weak crystalline orientation, as evidenced by the suppressed (222) diffraction peak intensity in XRD patterns. In contrast, higher substrate temperatures (> 200 °C) and oxygen partial pressures promoted enhanced crystallinity. High-quality amorphous films were achieved under low substrate temperature and oxygen partial pressure (< 4%), while polycrystalline films formed at elevated conditions. The crystallinity of ITO films was analyzed by X-ray diffraction (XRD, Bruker D8 Advance, Cu $K_{\alpha 1}$, $k = 1.54056$ Å). Optical transmittance in the UV-Vis-NIR range was measured using Shimadzu spectrophotometers (UV3600Plus and UV2700). Measurements were background-corrected using blank BaF_2 substrates processed under identical conditions. ITO thin film spectroscopic ellipsometric data were acquired using a J. A. Woollam Company Ellipsometer, model RC2 XI+. Spectroscopic ellipsometry data were analyzed using *CompleteEASE® Software (Version 6.70, J. A. Woollam Co., Inc., Lincoln, NE, USA*; available at https://www.jawoollam.com/ellipsometry-software/completeease. Spectroscopic ellipsometer data, including Psi and Delta, were recorded at an incidence angle of 60° across a spectral range of 250–936 nm (1.32–4.96 eV) in 1 nm increments. To account for surface and interface effects, very thin top and bottom layers were incorporated using a B-Spline model from the database, enhancing the realism and conformity of the fitting. Employing the Cody-Lorentz model for fitting, the main ITO layer thickness was determined to be approximately 412.5 nm, with the refractive index (n) and extinction coefficient (k) values presented in the subsequent section. Devices with Hall bars were fabricated on commercial silicon wafers with a 90 nm thermally grown SiO_2 layer. A 50 nm amorphous SiO_2 encapsulation layer was deposited at 80 °C by plasma-enhanced chemical vapor deposition (PECVD, Samco PD-220NL). Standard microfabrication steps were followed, including spin coating, baking, electron-beam lithography, development, dry etching, metallization, and lift-off. Electrical contacts were realized by sequential deposition of 30 nm Ni (electron-beam evaporation) and 30 nm Au (thermal evaporation), forming quasi-one-dimensional Ohmic contacts with contact resistances below 1 kΩ. All fabrication steps were carefully managed to minimize moisture exposure. Transport measurements were performed using a Quantum Design Physical Property Measurement System (PPMS) integrated with the dilution refrigerator insert. A constant-current mode was used with excitation currents between 50 and 2000 nA to ensure minimal self-heating and optimal signal-to-noise ratios. Magnetic properties were characterized using a Quantum Design Magnetic Property Measurement System (MPMS3) with Evercool technology for helium-free operation. The LLM (ChatGPT) was only used to polish the language in some parts of our manuscript.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Author contributions
K. W. fabricated devices. K. W., C. Z. and R. L. performed the measurements. K. W. and X. L. proposed the methodology and formal analysis. C. S. and X. L. provided the resources. K. W., X. S. and X. L. drafted, reviewed and edited the paper. This project was supervised by X. L. All authors reviewed the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

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